

Customer Shopping Behavior Analysis

Analyzing 3,900 transactions across product categories to uncover spending patterns, customer segments, and subscription behavior for strategic business decisions.



DATASET OVERVIEW

Inside the Dataset

3,900

Transactions

18

Columns

37

Missing Values

Key Features Tracked

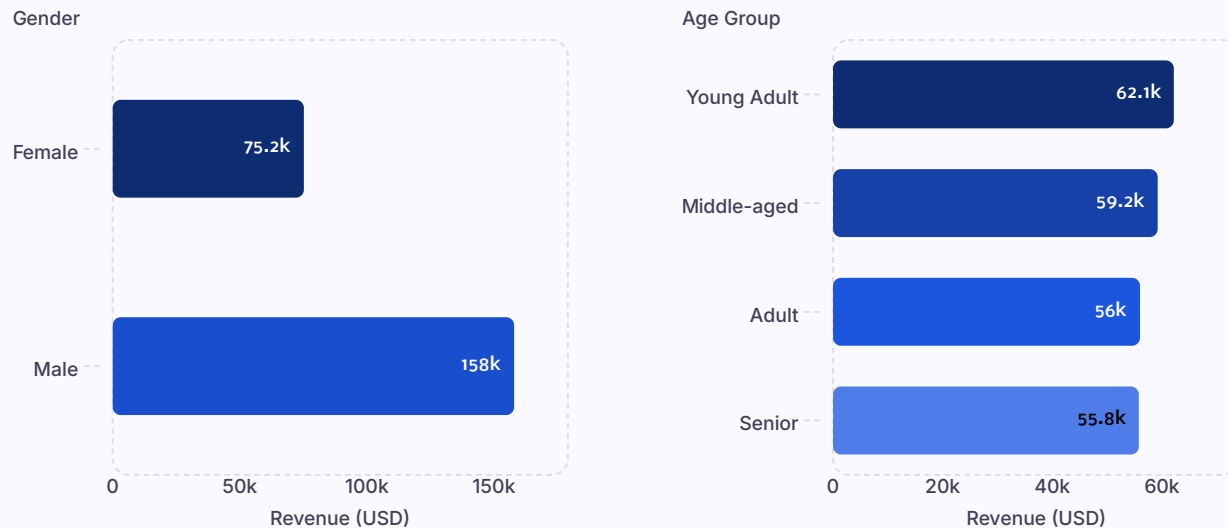
- Customer demographics: Age, Gender, Location, Subscription Status
- Purchase details: Item, Category, Amount, Season, Size, Color
- Shopping behavior: Discount, Promo Code, Previous Purchases, Frequency, Review Rating, Shipping Type

Data Preparation Pipeline



Columns standardized to snake case. Redundant promo_code_used column dropped after confirming it duplicated discount_applied. Cleaned DataFrame loaded into PostgreSQL for SQL analysis.

Revenue by Gender & Age



Male customers generate over **2x the revenue** of female customers.
Young Adults lead revenue contribution across all age groups.



Top Products & Discount Dependence

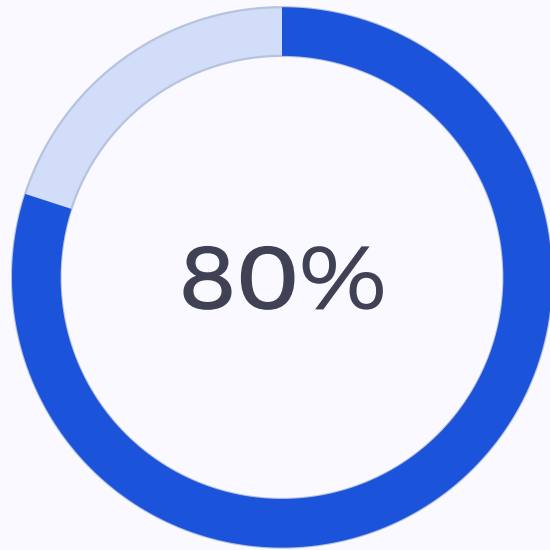
Top 5 Products by Rating

Product	Avg Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Most Discount-Dependent Products

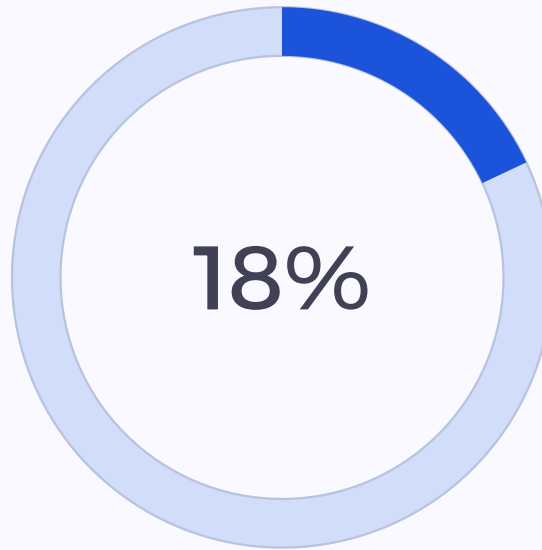
Product	Discount Rate
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

Customer Segmentation



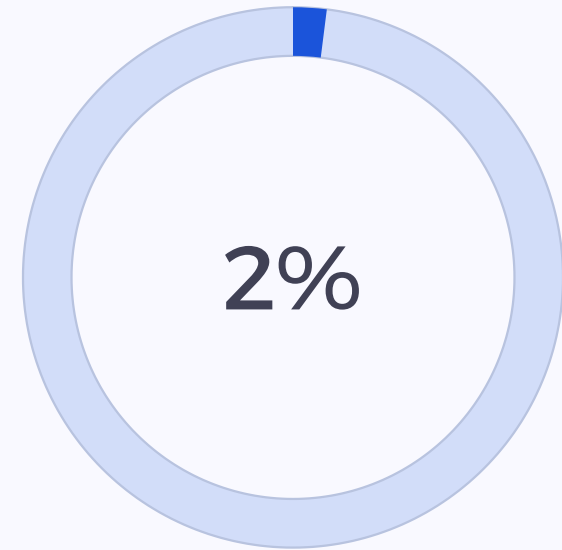
Loyal Customers

3,116 customers with strong purchase history



Returning

701 customers with moderate purchase activity



New

83 first-time customers entering the funnel

Among repeat buyers with more than 5 purchases, **2,518 remain non-subscribers** while only 958 subscribe – a major untapped conversion opportunity.

SQL ANALYSIS

Top Products per Category



Accessories

Jewelry (171), Sunglasses (161), Belt (161)



Clothing

Blouse (171), Pants (171), Shirt (169)



Footwear

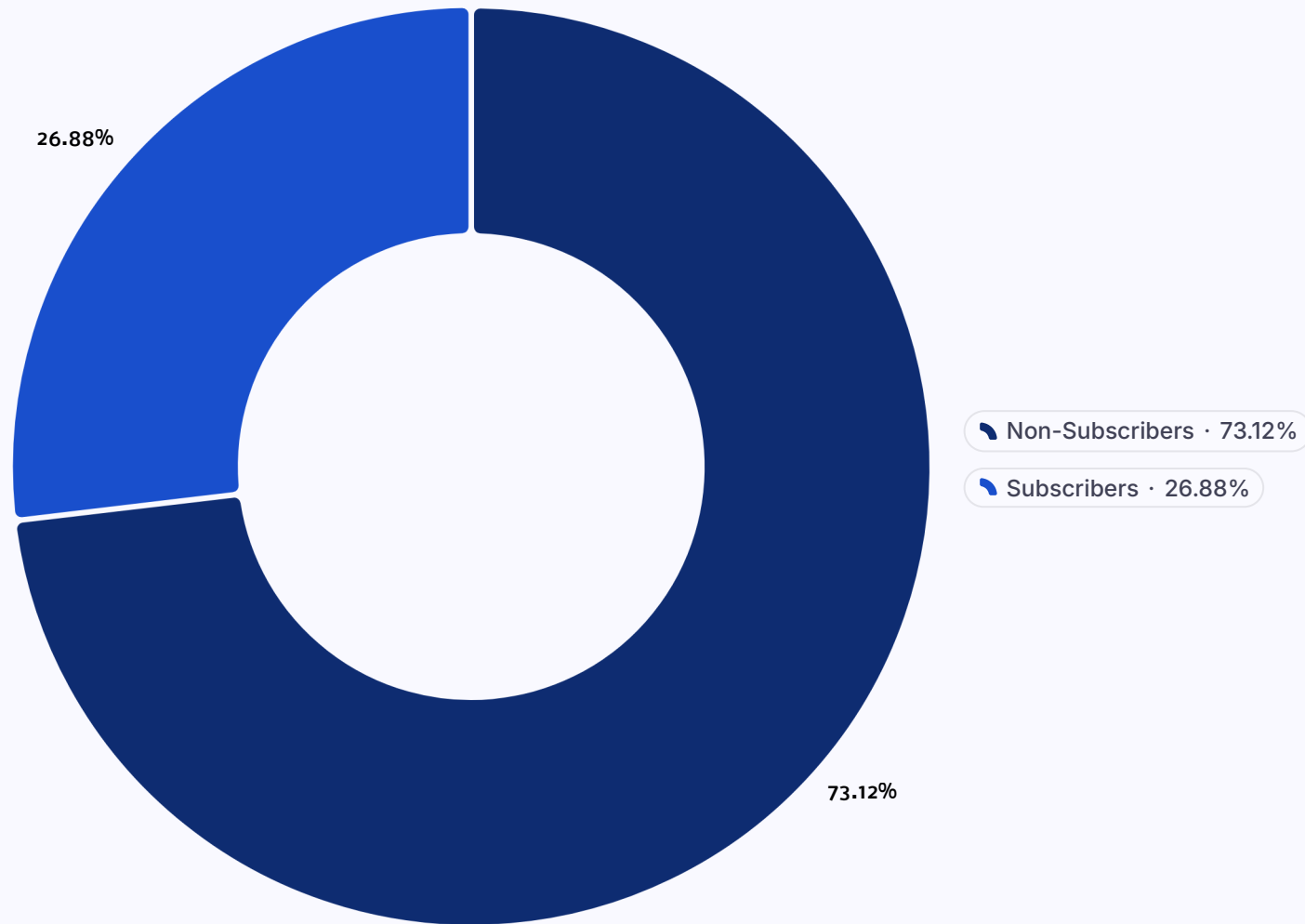
Sandals (160), Shoes (150), Sneakers (145)



Outerwear

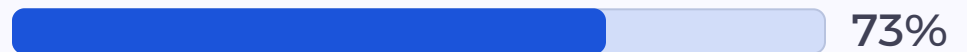
Jacket (163), Coat (161)

Subscribers vs. Non-Subscribers



Subscribers

1,053 customers, avg spend \$59.49



Non-Subscribers

2,847 customers, avg spend \$59.87

Average spend is nearly identical across both groups – subscription status doesn't currently drive higher individual purchase values.

POWER BI DASHBOARD

Interactive Dashboard Overview

3.9K

Customers

\$59.76

Avg Purchase

3.75

Avg Rating

Filters include subscription status, gender, category, and shipping type. Revenue by category shows Clothing leading at ~\$100K, followed by Accessories, Footwear, and Outerwear.



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RECOMMENDATIONS

Strategic Recommendations

Boost Subscriptions

Promote exclusive benefits to convert the 2,518 non-subscriber repeat buyers.

Loyalty Programs

Reward repeat buyers to move them into the Loyal segment.

Review Discount Policy

Balance sales boosts with margin control on discount-dependent products.

Product Positioning

Highlight top-rated and best-selling products in marketing campaigns.

Targeted Marketing

Focus on high-revenue age groups and express-shipping users.